UM-3 provenance

Where every input of UM-3 comes from, every decision taken, every computation and its checks, the external data and software, and how to reproduce all of it

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Provenance of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

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1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \ldots, x_7) :\ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The author's metric depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one surface of $\mathcal{F}$, at $z = \pi/2$: $\mathcal{F}$ is the union of these copies, one per period (example 2), and "the fold" without a qualifier is the copy in the period that contains our world. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4. Notation: ln is the natural logarithm, e Euler's number, $\exp(y) = \mathrm{e}^y$, sin, cos and cot = cos / sin the trigonometric functions, tanh the hyperbolic tangent, $|y|$ the absolute value; $f(0^+)$ and $f(0^-)$ are the limits of $f(\sigma)$ as σ approaches 0 from above and from below.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the author's metric is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2 \qquad \text{(FD-4)}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
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- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. The function $\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|}$, which multiplies the squared steps of ordinary space and of the extra times, is the warp factor of the metric (its square root $\mathrm{e}^{-H|\sigma|}$ multiplies the lengths); the time x_4 is not warped. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp factor is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$ equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\, g^{ac}\, \partial_\sigma g_{cb} : \qquad K^1{}_1 = K^2{}_2 = K^3{}_3 = K^5{}_5 = K^6{}_6 = K^7{}_7 = -H\,\mathrm{sgn}(\sigma), \qquad K^4{}_4 = 0 \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface $\sigma = \text{const}$ of the hidden direction, $K^a{}_b = \frac{1}{2} g^{ac} \partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along ordinary space and the extra times, 0 along the time. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8 X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

(a, b frame indices along the fold); it jumps by $-2H$ along ordinary space and along the extra times. The Einstein-Lovelock equations of UM-3, $\sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b + \Lambda \delta^a{}_b = \kappa \mathcal{T}^a{}_b$ ($\mathcal{E}_{(p)}$ the Lovelock tensor of order $p = 1, 2, 3$, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ its coupling with the weights $w = (1, 1/20, 1/360)$, $\Lambda = -7H^2$ the cosmological constant, $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions, $\mathcal{T}^a{}_b$ the energy-momentum tensor, $\delta^a{}_b$ the Kronecker delta), are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_\mathrm{D}(\sigma)$ into them: the metric solves them only if the fold carries a thin source $\mathcal{S}^a{}_b$, an energy-momentum tensor concentrated on $\sigma = 0$, given by the integral of the equations across the fold (the junction conditions),

$$\kappa\, \mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^a{}_b \tag{FD-11}$$

Symbols in (FD-11).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.

- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5} H \delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

with $\mathcal{S}^4{}_4 = -$(its energy density), $\mathcal{S}^1{}_1 = \mathcal{S}^2{}_2 = \mathcal{S}^3{}_3$ its pressure along ordinary space and $\mathcal{S}^5{}_5 = \mathcal{S}^6{}_6 = \mathcal{S}^7{}_7$ its pressure along the extra times; $\mathcal{S}$ is $\mathcal{T}$ integrated across the fold, an energy per unit volume of the six other directions of the fold (unit GeV^7). On the static block (a_4 and b_4 constant) and on an expanding fold whose extra times are frozen (b_4 proportional to e^{a_4}; $\mathsf{A} = a_4'/H$, $\mathsf{A}_1 = a_4''/H^2$)

$$\begin{aligned}
&\text{static block:} \quad \kappa\mathcal{S}^a{}_b = -\frac{32}{5} H\, \delta^a{}_b \quad \text{(a pure tension)},\\
&\text{frozen extra times:} \quad \kappa\mathcal{S}^1{}_1 = -\frac{16}{15}\big(6 + 3\mathsf{A}^2 + 2\mathsf{A}_1\big) H, \quad \kappa\mathcal{S}^4{}_4 = -\frac{16}{5}\big(2 + \mathsf{A}^2\big) H, \qquad \text{(FD-12)}\\
&\qquad\qquad \kappa\mathcal{S}^5{}_5 = -\frac{4}{5}\big(8 + 8\mathsf{A}^2 + \mathsf{A}^4 + 4\mathsf{A}_1 + \mathsf{A}^2\mathsf{A}_1\big) H
\end{aligned}$$

Symbols in (FD-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b$ (FD-11, the junction conditions); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -\frac{32}{5} H \delta^a{}_b$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); 1 at the author's vacuum, which is not a solution with the

corrected weights.

- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless.

the tension plus matter that obeys the Friedmann equations of general relativity. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0x_0} = \cot^2(6Hx_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2\Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \dots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-3 the Standard Model (the Higgs doublet with the author's ad hoc vacuum, a time-dependent choice of its vacuum value; the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$; three generations of quarks and leptons and three right-handed neutrinos) and the fold fluid (its tension and the pressure along the extra times that keeps their volume fixed) live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \dots, x_4$ (no field depends on the extra times). Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \dots, x_4$. dirac16complex (the author's spinor field of eight dimensions) and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the bound states χ of dirac16complex are the part of it that the fold holds, and the portals (couplings between dirac16complex and Standard-Model fields) act at $\sigma = 0$. In UM-3 files "the fold" means this surface; the examples below use the word for other surfaces with the same properties. The word is not used in its other senses: a set of equal energy levels is called degenerate, and a growth by the factor e is called an e-folding.

Examples of folds that are meaningful for UM-3.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the Standard Model, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp factor $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. A compact direction of length L would give every field a tower of particles with the masses $2\pi n/L$, $n = 1, 2, \dots$ (a Kaluza-Klein tower); because the copies are infinitely far apart, the period $\pi/(6H)$ of x_0 produces no such tower with the spacing $12H$, and the x_0 mass of the bound states is their binding to the fold.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property 3). In UM-3 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \dots, x_7$, and the crease is the fold.
4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_{\mathrm{D}}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\, \delta_{\mathrm{D}}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every metric function of FD-4 depends on σ through $|\sigma|$ and is folded at $\sigma = 0$ in this sense. 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_{\mathrm{RS}}^2 + \mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_{\mathrm{RS}} = \frac{6k_{\mathrm{RS}}}{\kappa_5}, \qquad \Lambda_{\mathrm{RS}} = -\frac{6k_{\mathrm{RS}}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y_{RS}: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-3). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_{RS}: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp factor of the brane is $\mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}$ (FD-8, an example, not UM-3). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_{RS}: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_{\mathrm{RS}} = 6k_{\mathrm{RS}}/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, $\kappa_5 = 8\pi G_5$ (FD-8, an example, not UM-3). Unit: GeV^{-3}. Status: STANDARD.
- Λ_{RS}: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_{\mathrm{RS}} = -6k_{\mathrm{RS}}^2/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^5. Status: STANDARD.

has a fold at $y_{\mathrm{RS}} = 0$ with the properties 1 to 3 (its coordinate y_{RS} is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_{RS} (Israel's junction condition) and in the bulk the vacuum energy Λ_{RS}, and the bulk is otherwise empty. The fold of UM-3 is of the same kind: its static block is a vacuum of the bulk with a fold of pure tension (FD-12). It differs in that the time x_4 is not warped, three extra times are warped together with ordinary space, the Lovelock terms add to the junction conditions, and the space is not cut at the brane: Randall and Sundrum identify the two sides of their brane (one physical side), while the fold of UM-3 has two distinct sides (example 7). 7. *A particle bound to a thin layer.* A particle held by an attractive thin layer has a wave function that falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-3 the particles χ of dark matter are the states of dirac16complex bound to the fold; at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega_{\mathrm{b}}x_4}$)

$$\chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+) \quad (\sigma > 0), \qquad \chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-) \quad (\sigma < 0),$$
$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}\big)\Big)\chi(0^-), \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0 T_1 \cdots T_7$ (the author's T16ˆA[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

where μ_L, μ_R are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the Standard-Model fields on the fold) and $q_\mathrm{b} = -m_\Psi(\mu_\mathrm{L} + \mu_\mathrm{R})\tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$ for $\mu_\mathrm{R} = 0$). The profile is not a function of $|\sigma|$: it jumps at the fold, $\chi(0^+) \neq \chi(0^-)$. 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0 T_1 T_2 T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0 T_5 T_6 T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0 T_1 T_2 T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the bound states of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0 T_5 T_6 T_7\,\chi(-\sigma)$ (FD-10): it maps the bound states of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of the mean field of a gas of the particles χ (the average field that the particles of the gas create and feel): the four bound states of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so the mean field of the gas cannot mix the two signs of the indefinite charge. R_2 maps the bound states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory (the eleven-dimensional theory whose limits include the superstring theories) on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); each carries one E_8 gauge group while gravity fills the bulk. As in UM-3, the gauge fields live on the mirror surface and gravity in the bulk; unlike UM-3, the two sides of each plane are identified (one physical side).

What is not a fold.

- *The tips* $z = 0$ and $z = \pi$ (mod π): the mirror exchanges them within a period, but there the warp factor vanishes, $g_{x_0x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp factor*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma = 0$, but without a kink and without a thin source.

2 How to read this document

This document records the origin of everything in the universe model UM-3: the author's inputs and orders, the decisions taken while building the model, the computations that verify every derived statement (each a Python script that checks itself), the external measurements and software that were used, and the commands that reproduce every result and every document. It is a record of origins, decisions and computations, not of the model itself, and so it contains no equations other than those of the definition of the fold at its beginning and the record of the stabiliser of the bulk (IN-7).

The status labels used throughout UM-3: **AUTHOR** (from the author), **RECORD** (from the record of this repository), **STANDARD** (textbook physics or a published measurement), **ASSUMED** (a choice without derivation), **INVENTED** (a new interaction: the portals of dirac16complex), **DERIVED** (computed and verified), **ESTIMATE** (an order of magnitude), **OPEN** (not settled).

3 The author's inputs

input	content and place
the order of 2026-10-24, 17:02 PDT	stop every earlier stage permanently; a new revision folder; read `4+4.nb` and verify its eight real 16×16 Dirac matrices; the new metric with b_4^2 (b_4 a function of the time x_4 in the extra-time components of the metric, never identically 1); the complete coupled field equations of the listed fields (Einstein-Lovelock gravity, the Higgs doublet with the ad hoc vacuum, three generations of lepton doublets, the right-handed electron, muon and tau, W^1, W^2, W^3, B, three generations of quark doublets, the gluons, dirac16complex coupled to gravity with a non-trivial spin connection and with its left- and right-handed halves coupled to every Standard-Model field, only the left-handed half to the Higgs field); new phenomenological interactions; an accurate approximate Kohn-Sham solution; the CMB, the CNB and the BAO with the metric as a power series; the x_0 mass mechanism; dark matter and dark energy; every equation and every symbol in every document, never a reference to another file; Markdown, LaTeX and PDF; push as soon as verified. In the author's private prompt file (not committed).
the notebook 4+4.nb	the author's notebook "eight, real-valued, 16 by 16, Dirac Matrices" (Wolfram 15.0) with the matrices `T16^A[0..8]`; copied unchanged into the folder `author_inputs` of UM-3 (its sha256 is given below)

input	content and place
the notebook of the record	`Dirac_Einstein-Lovelock_4+4_Nash.nb` in the root of the repository: the coordinates $X = \{x_0, \ldots, x_7\}$ (Section DEFCHART: x_0 the hidden direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra times) and the Lovelock weights $w = (1, 1/32, 1/1152)$ of the Einstein, Gauss-Bonnet and cubic terms with the cosmological-constant parameter $\Lambda_{\mathrm{N}} = 10$ of the notebook ($\Lambda = -10H^2$; In[134]); corrected at the order of 2026-10-25, 07:55, to $w = (1, 1/20, 1/360)$ and $\Lambda = -7H^2$
the decision of 17:32	"rather have three ν_{R}": three right-handed neutrinos, Dirac masses (in the conversation)
the reason of 17:34	Gogilashvili and Tamborra, "Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504: why the flavour conversion of UM-3 must be correct (a third-party paper on the author's computer, read, not committed)
the standing rule of 2026-10-14	no single calculation may take longer than 423 s (7 min 3 s); every command runs under a time limit (recorded in `claude_failures.md` and `HANDOFF.md` of the repository)
the order of 2026-10-25, 03:21 PDT	handle the sources on the fold: the fold needs a thin fluid and an Einstein-Hilbert term of its own (without the latter, bulk gravitons would travel 2.74 times faster than light); push. The figure 2.74 restated the earlier documents of UM-3, in which it was a deduction of Claude from the author's original weights (the long-wavelength wave of the bulk alone, $c_{\mathrm{g}}^2 = 15/2$), not an assumption of the author
the order of 2026-10-25, 03:23 PDT	analyse and answer the open questions: the stability of the indefinite Krein norm of the charge of dirac16complex; fields that depend on the extra times (tachyonic); superluminal particles χ unless the binding is tight ($q_{\mathrm{b}} \gg H$); the primordial era, for which the Planck values were inputs; the lightest neutrino mass and the Dirac phase, which had been set to zero
the order of 2026-10-25, 03:54 PDT	define the fold clearly, correctly and precisely, with as many examples meaningful for UM-3 as possible, and present the same definition and examples in every file that uses the word, never referring to another file
the order of 2026-10-25, 05:09 PDT	give proper attribution to Claude for the stabiliser; discuss it in its own complete documentation with provenance (Markdown, LaTeX, PDF) for an ignorant but clever student of science; verify that it couples only to gravity, that it is absent in the primordial era, and that its existence and value are derived from the requirement of a static bulk; explain the static bulk with examples; define warp and warp factor; analyse its importance for UM-3
the order of 2026-10-25, 07:55 PDT	in reply to the statement that bulk gravitons still travel at $\sqrt{5}$ times the speed of light and that the 2.74 came from leaving out the fold's own Lovelock term: "I do not understand … because I did not presume 2.74 c anywhere. I assume this is a deduction from other assumptions. Please explain. This needs to be fixed because Gravitons must travel at c. If these incorrect claims come from my choice of the Lovelock couplings and my choice of the cosmological constant Λ THEN my choices are wrong, and you MUST correct them." Claude explained that the figure was its own deduction from the original weights, and replaced the weights and Λ throughout UM-3 by the only values for which gravitons travel at the speed of light

The sha256 of the author's notebook `4+4.nb` as received and committed:

`7ec1709a0028bce0b443f116cf27b5d3e670de7aa195659a9c3274527270429d`

The author's private files (his prompt files, his notes, the third-party paper, his scratch notebook of 2026-10-16 with its `.mx` files and the files `Generalized_Kronecker_Delta.*`) are never committed; the repository's `.gitignore` excludes them.

4 The decisions taken in UM-3

decision	reason	status
the proper coordinate σ of the hidden direction; the fold at $z = \pi/2$ ($z = 6Hx_0$, the angle on which the metric depends)	the author's metric has a kink in σ there, which the junction conditions turn into a thin source	DERIVED
the Standard Model on the fold, dirac16complex in the bulk	species in the bulk have different limiting speeds and neutrino profiles would spoil flavour conversion	ASSUMED (motivated by the DERIVED limiting speeds)
right-handed quark singlets and Yukawa couplings	masses of the quarks and cancellation of the anomalies (defaults of the notification of 17:20; the author answered only on the neutrinos)	ASSUMED
three right-handed neutrinos	the author's decision of 17:32	AUTHOR
the Lovelock weights $1, 1/20, 1/360$ and $\Lambda = -7H^2$ (the order of 2026-10-25, 07:55)	gravitons travel at the speed of light in the bulk and on the fold only if both coefficients of the bulk wave, $1 - 40w_{(2)} + 360w_{(3)}$ and $1 - 24w_{(2)} + 72w_{(3)}$, vanish: the unique solution; with it the static block is a vacuum only for $\Lambda = -7H^2$	DERIVED (`um3_weights.py`)
the gravity of the fold	with the corrected weights the Gauss-Bonnet and cubic terms give the fold its own Einstein-Hilbert term, with the Friedmann equations of general relativity and luminal gravitational waves; an induced term of the fields of the fold is optional (with the original weights it had to provide all but 1.3×10^{-13} of the gravity of the fold, because the waves moved with $c_g^2 = 83/82$)	DERIVED; ESTIMATE (the induced term)
the fold fluid	the kink at the fold needs a thin source: a pure tension at the static block, and a pressure q_{hold} along the extra times that holds their volume fixed (a free volume would change Newton's constant far beyond the bounds of lunar laser ranging, optical clocks, Oklo and nucleosynthesis)	DERIVED (its values), ASSUMED (the medium and its holding pressure), OPEN (the origin of the holding pressure)
the stabiliser of the bulk (a record)	with the original weights the static bulk needed exactly the stabiliser, which Claude found and named on 2026-10-24; with the corrected weights it vanishes, and with it the bulk media of the expanding fold and the radion mode that the earlier versions assumed	DERIVED (its value and its removal)

decision	reason	status
the portals of dirac16complex	the author's request for couplings of both halves to every Standard-Model field, only the left-handed half to the Higgs field	INVENTED
the pure-gauge W^2 of the ad hoc vacuum	the Gauss law of $SU(2)$	DERIVED
no dependence of any field on the extra times	such fields are tachyonic; the restriction is consistent but unstable, and a gauge principle that removes the momentum along the extra times is needed	ASSUMED (OPEN: the gauge principle)
the lightest neutrino mass zero, the Dirac phase zero	not fixed by the specification; the phase drops out of every result of UM-3, and DESI DR2 with the CMB and NuFIT 6.0 bounds the lightest mass below 23 meV	ASSUMED (the mass within the bound), DERIVED (the phase is irrelevant)
the primordial spectrum, the optical depth and the densities from Planck 2018	the author's vacuum is not a solution with the corrected weights; the primordial era is an inflation of the fold with the volume of the extra times held fixed, the inflation of general relativity, whose inflaton UM-3 does not specify (the Higgs field is possible)	ASSUMED (inputs; the inflaton OPEN)
the quantisation of dirac16complex with positive energies	it makes the vacuum stable; the modes of positive Krein charge are then ghosts, which the chiral couplings of the fold produce; the classical stability of the gas is not decided	DERIVED (the consequences), OPEN (the stability of the gas, a unitary coupling)
one definition of the fold in every file	the order of 2026-10-25, 03:54: the definition with its examples, `docs_src/fold_definition.md` for the documents and `docs_src/fold_definition.txt` for the code and the JSON files, inserted and checked by `tools/fold_block.py`	AUTHOR (the order)
a document of its own for the stabiliser of the bulk, `STABILISER.md`	the order of 2026-10-25, 05:09: the stabiliser was found, introduced and named by Claude (Anthropic) on 2026-10-24 (the gravity module, check G4, and the term IN-7 of the field equations); the document explains it for a newcomer, verifies the five points of the order with the checks of `um3_stabiliser.py`, records its provenance, and, after the order of 07:55, states that the stabiliser was a consequence of the author's original weights and vanishes with the corrected ones (check ST9)	AUTHOR (the orders)

decision	reason	status
the Kohn-Sham module in parts	the standing rule of 423 s: the module runs in six parts with snapshots, each under the time limit, and an assembly that reproduces the single run exactly	AUTHOR (the rule)

The lineage of the stabiliser of the bulk. Something that holds a static block of the bulk appears in the frozen records of this repository before UM-3: the first universe model (the folder `Revision/universe_model`, version 1.1, already at the first commit of the repository, `0fdf375`) assumed a "block stabiliser", a potential for the size $\mathrm{e}^{-a_4} b_4^{1/2}$ of the block of the hidden direction and the extra times of its metric (whose extra-time components carried b_4 instead of b_4^2), with an origin left open; the second (`universe_model_2`, version 3 of 2026-10-18) a block stabiliser whose energy density falls as the inverse eighth power of the block scale, $\kappa\rho_{\mathrm{stab}} = (26 + 4\sqrt{51})H^2 = 54.566H^2$ at its static point (ASSUMED there). The holding pressure q_{hold} of the fold fluid of UM-3 plays a similar part for the volume of the extra times. For the author's new metric of 2026-10-24 Claude derived the exact source that the static bulk requires (the gravity module, check G4, commit `4448938`), introduced it into the field equations of UM-3 as the stabiliser of the bulk, IN-7 (commit `170d8d3`), and described it in its own document (commit `cd5709d`). At the order of 2026-10-25, 07:55, the author's Lovelock weights and cosmological constant were replaced by the only values for which gravitons travel at the speed of light; with them the static bulk is a vacuum and the stabiliser vanishes (`um3_stabiliser.py`, check ST9). Its record and its removal, in the order of the directions σ, x_1, x_2, x_3, x_4, x_5, x_6, x_7:

$$\kappa\,\mathcal{T}_{\mathrm{stab}}{}^{a}{}_{b} = \Big(\sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^{a}{}_{b} + \Lambda\,\delta^{a}{}_{b}\Big)\Big|_{\text{static block}} = 0 : \qquad \text{(IN-7)}$$

$$(15 - 9 + 1 - 7)H^2 = 0 \text{ (warped directions)}, \qquad (21 - 21 + 7 - 7)H^2 = 0 \text{ (the time)}$$

Symbols in (IN-7).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{stab}}{}^{a}{}_{b}$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric required with the author's original Lovelock weights, found and named by Claude (Anthropic) on 2026-10-24; with the corrected weights and $\Lambda = -7H^2$ it is zero (IN-7). Unit: GeV^8. Value: 0 with the corrected weights. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

5 The computations

Every derived statement rests on one of these scripts in `universe_model_3/code/`. Each computes, checks itself and writes its results to `universe_model_3/outputs/` and its checks to `universe_model_3/reports/`.

script	what it computes	checks
`um3_gammas.py`	the author's Dirac matrices, rebuilt from `4+4.nb` (a library)	
`um3_dirac_matrices.py`	reality, Clifford relations, Spin(4,4) and Pin(4,4), chiral projectors, agreement with every stored output of the notebook	55
`um3_dirac_matrices_doc.py`	the document `dirac matrices` from the checked results	
`um3_lovelock.py`	the Lovelock tensors by the generalised Kronecker delta (a library)	
`um3_geometry.py`	the exact curvature and Einstein-Lovelock equations of the author's metric with the corrected weights, the vacua of the bulk, the two constraints, the author's vacuum (no longer a solution), the spin connection	23
`um3_fold.py`	the proper chart, the general warped metric, the junction potential and the fold sources, the conservation law at the fold, the Friedmann equations of the frozen branch	13
`um3_weights.py`	the bulk graviton and the graviton of the fold for any Lovelock weights, the unique weights for which gravitons travel at the speed of light, the cosmological constant of the static block, the vacua and the fold of the corrected weights, the record of the author's weights	8
`um3_bulk.py`	the zero modes and the speeds of the bulk fields, the principal symbol of the bulk graviton	11
`um3_fields.py`	the bilinears of dirac16complex, the Krein operator, the electroweak vacuum with the ad hoc component, the anomalies	16
`um3_spectra.py`	the bound states of dirac16complex with the exact matrices	5
`um3_neutrino.py`	the flavour blindness of the portals, the matter conversion, the crossing criterion, the equipartition, the cosmic neutrino background (CNB)	5

script	what it computes	checks
`um3_background.py`	the bulk and fold equations against the stored outputs, the two vacuum branches, the sources off the branches, the primordial era, the fold factor, the ad hoc vacuum as a gauge rotation	7
`um3_series.py`	the degenerate static block and its second-order operator, the junction of static deformations, the fold as a seven-dimensional Einstein brane, the volume of the extra times (free: excluded; held: general relativity)	6
`um3_kohnsham.py`	the self-consistent Kohn-Sham states of dirac16complex in three regimes and at a weakly bound point	9
`um3_flavour.py`	the neutrino flavours in the Sun and in supernovae	4
`um3_thermal.py`	the degrees of freedom, the relic density, the portals and direct detection, the right-handed neutrinos	6
`um3_cosmology.py`	the cosmic microwave background (CMB), the CNB and the baryon acoustic oscillations (BAO) with the Boltzmann code CAMB and an independent background, against Planck 2018 and DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument)	6
`um3_foldsources.py`	the gravitational wave of the fold: no bulk graviton, its thin part from the Lovelock terms, its speed (that of light at every wavelength), the induced term; the records of the author's weights (83/82, the dispersive bound mode, the GW170817 bound); the gravity induced by the fields of the fold; the fold fluid as a tension	6
`um3_stabiliser.py`	the record of the stabiliser of the bulk with the author's original weights (the source of the static bulk, its absence at the author's vacuum, its conservation, energy conditions and ghost field, history, energy balance, importance, the constraints) and its removal by the corrected weights	9

script	what it computes	checks
`um3_krein.py`	the Krein form and its quantisation, the symmetries and mirrors of the bound states, the flat directions of the gas and its truncated response, the partner level, the selection rules at the fold, stability and unitarity (reads the snapshots of `um3_kohnsham.py`)	7
`um3_extratimes.py`	fields with a momentum along the extra times: the dispersion, the bound states and their leaking, the growth on the frozen branch and in the author's vacuum (a record), compact extra times and their harmonics, the truncation	5
`um3_superluminal.py`	the speed coefficient of the particles χ, their dispersion up to large momenta, the speeds of cosmological χ, the closed time-like curves and the shortcut through the bulk, the loop estimate for the electron	5
`um3_primordial.py`	the author's vacuum (no longer a solution), the records of the fields of the fold in it, the inflation of the fold with the volume held and its tensor bound, Higgs inflation, the inflation with a free volume	7
`um3_numass.py`	the lightest neutrino mass and the Dirac phase: the masses of both orderings, the bounds of DESI DR2 and KATRIN, the phase in the results of UM-3, the effect of the lightest mass, the record against NuFIT 6.0	5

The documents are written in Markdown and built with `universe_model_3/tools/build_docs.py` (pandoc to LaTeX, pdflatex to PDF, deterministic). `universe_model_3/docs_src/registry.py` holds the meaning, unit, value and status of every symbol; `universe_model_3/tools/make_inventory.py` writes the inventory of the tagged equations from `FIELD_EQUATIONS.md`, and `universe_model_3/tools/symbols_tool.py` copies every tagged equation into the other documents (`sync`), writes the symbol list after every display (`expand`) and checks that every symbol of every display is defined and every tagged equation is written identically in every document (`check`). `universe_model_3/tools/fold_block.py` inserts the one definition of the fold into every file that uses the word (`insert`) and checks that every such file carries it word for word (`check`).

6 External measurements and software

source	used for
Planck Collaboration, Planck 2018 results VI, A&A 641, A6 (2020), Table 2 (TT,TE,EE+lowE+lensing)	the cosmological parameters and the derived quantities of the CMB
Planck 2018 plik-lite likelihood files (Planck Legacy Archive), via github.com/heatherprince/planck-lite-py	the CMB spectra (downloaded at run time into `universe_model_3/external/`, not redistributed)

source	used for
DESI Collaboration, DESI DR2 Results II, arXiv:2503.14738, Table IV	the BAO distances; the 2.3 and 3.1 standard-deviation statements of its abstract
ACT Collaboration (Atacama Cosmology Telescope), DR6 constraints on extended cosmological models, arXiv:2503.14454	$N_{\rm eff} = 2.86 \pm 0.13$ and 2.89 ± 0.11, $\sum m_\nu < 0.082$ eV
Borexino Collaboration, Nature 562, 505 (2018)	the survival probabilities of solar neutrinos
the global fit NuFIT 5.2 of the neutrino oscillation data (Esteban, Gonzalez-Garcia, Maltoni, Schwetz and Zhou, JHEP 09 (2020) 178; version 5.2 of November 2022 at www.nu-fit.org), normal ordering, the fit without the Super-Kamiokande atmospheric data	$\sin^2\theta_{12} = 0.303$, $\sin^2\theta_{13} = 0.02203$, $\sin^2\theta_{23} = 0.572$, $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.511 \times 10^{-3}$ eV^2
Particle Data Group, Review of Particle Physics (2022)	the masses of the Higgs boson (125.25 GeV), of the quarks and charged leptons and of the nucleon, the Fermi constant 1.1663788×10^{-5} GeV^{-2} and $v = 246.22$ GeV, the fine-structure constant
W. Elbers et al. (DESI Collaboration), arXiv:2503.14744, section IV C	DESI DR2 with the CMB combined with NuFIT 6.0 (95 per cent, cosmological-constant model): the lightest mass below 0.023 eV (normal ordering) and 0.024 eV (inverted), the sum below 0.101 eV and 0.133 eV, the probability 0.91 of the normal ordering; the bound 0.0642 eV on the sum of three equal masses
LUX-ZEPLIN (LZ) Collaboration, Phys. Rev. Lett. 135, 011802 (2025), arXiv:2410.17036 version 3 (4.2 tonne-years)	the limit 2.2×10^{-48} cm^2 at 40 GeV (90% confidence) on spin-independent scattering
Long, Lunardini and Sabancilar, arXiv:1405.7654	the capture rate of the CNB on tritium (4 and 8 per year per 100 g for Dirac and Majorana neutrinos)
Gogilashvili and Tamborra, arXiv:2605.16504 (version 3)	the flavour equipartition (the equal sharing of the three flavours that fast flavour conversion produces) and its effect on the explosions of massive stars
Bahcall's exponential solar density profile; Petcov's crossing probability; Keil, Raffelt and Janka's spectral shape; Izaguirre, Raffelt and Tamborra's dispersion relation; Morinaga's crossing theorem; Ma and Bertschinger's Boltzmann equations	the standard physics named in the documents
CAMB 2.0.5 (Lewis, Challinor and Lasenby)	the Boltzmann hierarchy and the CMB spectra
the global fit NuFIT 6.0 (I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro and T. Schwetz, JHEP 12 (2024) 216, arXiv:2410.05380), data set IC24 with the Super-Kamiokande atmospheric data, and IC19 without it ($\sin^2\theta_{12} = 0.307$, $\sin^2\theta_{23} = 0.561$, $\sin^2\theta_{13} = 0.02195$, $\delta_{\rm CP} = 177°$, $\Delta m^2_{31} = 2.534 \times 10^{-3}$ eV^2)	$\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\rm CP} = 212°$, $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2 (normal ordering), and the inverted ordering, for the comparison with the record
KATRIN Collaboration, Science 388 (2025) 180	the limit 0.45 eV (90%) on the effective electron-neutrino mass of tritium beta decay
M. Tristram et al., Phys. Rev. D 105, 083524 (2022) (Planck PR4 with BICEP/Keck 2018 and BAO)	the bound $r < 0.032$ (95%) on the tensor-to-scalar ratio
Planck Collaboration, Planck 2018 results X (A&A 641, A10, 2020); T. Louis et al. (ACT Collaboration), arXiv:2503.14452	the spectral index $n_{\rm S} = 0.9649 \pm 0.0042$ (Planck 2018) and 0.974 ± 0.003 (ACT DR6 with Planck, CMB lensing and DESI)
F. Bezrukov and M. Shaposhnikov, Phys. Lett. B 659, 703 (2008)	Higgs inflation

source	used for
LIGO Scientific, Virgo, Fermi GBM and INTEGRAL Collaborations, Astrophys. J. Lett. 848, L13 (2017)	the speed of gravitational waves, $-3 \times 10^{-15} < c_{\mathrm{g}} - 1 < 7 \times 10^{-16}$ (GW170817 and GRB 170817A)
A. D. Sakharov, Dokl. Akad. Nauk SSSR 177, 70 (1967); D. V. Vassilevich, Phys. Rep. 388, 279 (2003)	induced gravity; the heat-kernel coefficients of scalars, fermions and gauge fields
F. W. Stecker, Astropart. Phys. 56, 16 (2014), arXiv:1306.6095; T. Jacobson, S. Liberati and D. Mattingly, Nature 424, 1019 (2003)	the bounds on the maximal speed of electrons: faster than light by less than 5×10^{-21} (no vacuum Cherenkov radiation of the 5 PeV electrons of the Crab flare of 2010), slower by less than 5.6×10^{-20} (synchrotron electrons with the Lorentz factor 3×10^{9} in the Crab nebula)
L. Biskupek, J. Müller and J.-M. Torre, Universe 7, 34 (2021); M. Filzinger et al., Phys. Rev. Lett. 130, 253001 (2023); C. J. Copi, A. N. Davis and L. M. Krauss, Phys. Rev. Lett. 92, 171301 (2004)	the bounds on the change of Newton's constant (lunar laser ranging), of the fine-structure constant (optical clocks), and Newton's constant at nucleosynthesis, against a free volume of the extra times
S. C. Davis, Phys. Rev. D 67, 024030 (2003), arXiv:hep-th/0208205; E. Gravanis and S. Willison, Phys. Lett. B 562, 118 (2003)	the junction conditions of Lovelock gravity and the Einstein-Hilbert term that the Gauss-Bonnet term gives a brane
A. I. Shlyakhter, Nature 264, 340 (1976); T. Damour and F. Dyson, Nucl. Phys. B 480, 37 (1996)	the bound of the natural reactor of Oklo on the change of the fine-structure constant over 1.8 Gyr
S. Dubovsky, T. Grégoire, A. Nicolis and R. Rattazzi, JHEP 03 (2006) 025; J. M. Cline, S. Jeon and G. D. Moore, Phys. Rev. D 70, 043543 (2004)	the instabilities of media that violate the null energy condition, and of ghosts
A. Einstein (1917); A. S. Eddington (1930); K. Schwarzschild (1916); R. V. Pound and G. A. Rebka (1960); D. Lovelock (1971); L. Randall and R. Sundrum (1999)	the examples and the standard physics of the document on the stabiliser
Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath, matplotlib 3.11.2; pandoc 3.1.3; pdfTeX (TeX Live 2023)	every computation and document

7 How to reproduce

From the root of a clone of the repository, with the software above installed:

```
python universe_model_3/code/um3_dirac_matrices.py
python universe_model_3/code/um3_geometry.py
python universe_model_3/code/um3_fold.py
python universe_model_3/code/um3_weights.py
python universe_model_3/code/um3_bulk.py
python universe_model_3/code/um3_fields.py
python universe_model_3/code/um3_spectra.py
python universe_model_3/code/um3_neutrino.py
python universe_model_3/code/um3_background.py
python universe_model_3/code/um3_series.py
python universe_model_3/code/um3_flavour.py
python universe_model_3/code/um3_thermal.py
python universe_model_3/code/um3_cosmology.py
python universe_model_3/code/um3_kohnsham.py part K1
python universe_model_3/code/um3_kohnsham.py part I
python universe_model_3/code/um3_kohnsham.py part II
python universe_model_3/code/um3_kohnsham.py part III
python universe_model_3/code/um3_kohnsham.py part K6
```

```
python universe_model_3/code/um3_kohnsham.py part K9
python universe_model_3/code/um3_kohnsham.py assemble
python universe_model_3/code/um3_foldsources.py
python universe_model_3/code/um3_stabiliser.py
python universe_model_3/code/um3_krein.py
python universe_model_3/code/um3_extratimes.py
python universe_model_3/code/um3_superluminal.py
python universe_model_3/code/um3_primordial.py
python universe_model_3/code/um3_numass.py
python universe_model_3/docs_src/registry.py
python universe_model_3/tools/make_inventory.py
python universe_model_3/tools/symbols_tool.py sync
python universe_model_3/tools/fold_block.py insert
python universe_model_3/tools/symbols_tool.py check
python universe_model_3/tools/fold_block.py check
python universe_model_3/tools/build_docs.py
```

Every script prints one line per check and ends with "n/n checks passed"; `um3_series.py`, `um3_background.py` and `um3_foldsources.py` read the outputs of `um3_fold.py` (and `um3_background.py` and `um3_primordial.py` that of `um3_geometry.py`), `um3_bulk.py` reads the report of `um3_weights.py`, `um3_primordial.py` the output of `um3_background.py`, `um3_krein.py` reads the snapshots of the parts of `um3_kohnsham.py`, and `um3_cosmology.py` downloads the Planck files once (about 7 MB). By the standing rule no single calculation may take longer than 423 seconds: each command is run under `timeout 423`; a part of `um3_kohnsham.py` that reaches its own limit of 400 seconds saves its state and exits with the code 3, and is then simply run again until it exits with 0 (the six parts together take about 37 minutes of computing; the longest single module otherwise is `um3_krein.py`, about 4.5 minutes). The documents are rebuilt identically (no dates in the LaTeX, fixed PDF metadata).